GPS receivers, GPS deselection and ECAM STATUS

Real GPS receivers in MMR 1 and 2 for both aircraft, with the GPIRS position, GPS PRIMARY from the real accuracy, the GPS pages, the NAV GPS FAULT alerts, GPS deselection and the A32NX ECAM STATUS page; then the FCOM FM position (IRS plus a GPS or radio bias) with the POSITION MONITOR pages and the FCOM FMS messages

Aircraft	A32NX and A380X (shared systems code); the ECAM STATUS page and STS reminder are A32NX only
Where to find it	MCDU DATA > GPS MONITOR and SELECTED NAVAIDS (A32NX); MFD FMS > POSITION > GPS and NAVAIDS (A380X); E/WD and SD STATUS page; flyPad > Failures > ATA 34 Navigation; MCDU POSITION MONITOR, PROG and messages; MFD POSITION / MONITOR and FMS messages
Branch	`core/gps`, based on `core/a380x-mfd-fms` (stacked: needs core/surveillance and core/a380x-mfd-fms merged first). `core/a32nx-air-conditioning`, `core/a380x-elec` and `core/failures-a32nx` build on it.
Status	All six features tested in the sim by the user: OK: receivers and pages, A380X DIFF mode, GPS deselection on both aircraft, NAV GPS FAULT alerts, A32NX STATUS page and STS reminder, A380X INOP SYS (on a combined test build). Not sim-tested on this branch: the A32NX STS reminder, adapted here to the FlyByWire master CLR logic. A full MSFS restart is needed after installing (systems WASM). FM position and FMS messages: not sim-tested, unit tests only.
Pictures	One picture: the A380X POSITION/GPS page layout, taken in the browser harness before the receivers existed (the values on it are the old stand-in values). The other displays are text only.

1. GPS receivers in MMR 1 and 2
2. GPIRS position and GPS PRIMARY
3. GPS pages: MCDU GPS MONITOR and MFD POSITION/GPS
4. NAV GPS 1(2) FAULT alerts
5. GPS deselection and position uncertainty
6. A32NX ECAM STATUS page and STS reminder
7. FM position: IRS position plus a GPS or radio bias, POSITION MONITOR pages
8. FMS messages from the FCOM lists, and CHECK APPR SELECTION

1. GPS receivers in MMR 1 and 2

> **Aircraft:** A32NX and A380X | **Where:** No control: the receivers run when powered | **Status:** Tested in the sim by the user: OK. Degraded modes also covered by 21 Rust unit tests.

What it is for

There was no GPS receiver in either aircraft: the pages showed the sim GPS position for both receivers with random satellite counts and figures of merit, and GPS PRIMARY came from a random estimate. Now each Multi-Mode Receiver (MMR 1 and MMR 2) has a GPS receiver. A320 FCOM DSC-34: "Worldwide, 24 satellites broadcast accurate navigation data"; the two independent receivers process them and send their data to the ADIRUs.

- **Satellites:** the sim has none, so they are computed from the nominal 24-slot GPS constellation (six orbital planes, four satellites each) at the sim UTC date and time. A satellite counts when it is more than 5° above the horizon. Their geometry gives the dilution of precision, and from it the figure of merit (accuracy) and the integrity limit (HIL) of the position. Fewer than 5 satellites: no integrity limit.
- **Modes** (A320 FCOM DSC-22_20 GPS MONITOR, A380 FCOM DSC-34-10-40-10): **TEST** (self test at power up, 1.5 to 2.5 s), **INIT** (2.5 to 4.5 s), **ACQ** (satellite acquisition), **NAV** (4 or more satellites, data to the ADIRUs), **FAULT** (failed receiver).
- **Acquisition time:** 15 to 30 s after a short power cut (unpowered 15 min or less: warm start), 80 to 130 s from a cold start, 5 to 15 s to re-acquire after a loss of the signals.
- **Degraded modes** (the FCOMs only name them; the behaviour follows ARINC 743A, a design choice): **ALTAID** with 3 satellites (the ADR barometric altitude is the fourth measurement, no integrity limit); **AIDED** with fewer satellites,

the receiver coasts on the IR velocities for up to 2 min, its accuracy gets worse with time and it gives no integrity limit. After that: back to ACQ.

- **Antenna masking:** the antenna is on top of the fuselage. In a turn or a steep climb, satellites low on the far side of the aircraft are masked.
- **DIFF (A380X only):** the A380 receivers use satellite based augmentation. In NAV inside a WAAS (North America), EGNOS (Europe), MSAS (Japan) or GAGAN (India) service area, the mode shows DIFF and the position is several times more accurate. The service areas are coarse latitude/longitude boxes (design choice).
- **Power:** A32NX: the FCOM gives none. GPS 1 is on AC ESS and GPS 2 on AC 2, as the other side 1 / side 2 navigation receivers (design choice). A380X: as the FCOM supply table, GPS 1 (MMR 1) is on AC EMER and GPS 2 (MMR 2) on AC 3. The bus named AC ESS in the A380X code is the AC EMER busbar.
- **Failures:** flyPad > Failures > ATA 34 Navigation > **GPS 1** and **GPS 2** put the receiver in FAULT.

Limitations and why

Limitation	Why	What to do
The satellite positions are the nominal constellation, not the real satellites of the day.	No almanac file is used: the 24 nominal slots give a realistic count and geometry anywhere in the world.	None needed.
ALTAID and AIDED are rarely seen in flight.	With 24 satellites, 4 or more are almost always in view; masking in a steep turn seldom leaves fewer.	Covered by the Rust unit tests.
A full MSFS restart is needed after installing.	The receivers are in the systems WASM module.	Restart the sim.

How to use

1. Nothing to set: power the aircraft and the receivers start (TEST, INIT, ACQ, then NAV).
2. Watch them on the A32NX MCDU **DATA > GPS MONITOR** page or the A380X MFD **FMS > POSITION > GPS** page (see "GPS pages" below).

How to test

#	Do this	Expected result
1	Cold and dark (aircraft unpowered more than 15 min). Power the aircraft (batteries, external power or APU) and open the GPS page at once.	Both receivers show TEST briefly, then INIT, then ACQ with a rising number of satellites; NAV after about 80 to 130 s.
2	Remove the power for less than 15 min, then power again.	ACQ lasts only 15 to 30 s (warm start).
3	In NAV, read the satellites and the figure of merit.	Between 4 and 12 satellites (7 to 10 on average) and a merit of about 30 to 100 ft; the two receivers slightly different.
4	A380X in Europe or North America (for example EGLL or KJFK): open POSITION / GPS.	MODE shows **DIFF** on both receivers.
5	A380X outside the service areas (for example SBGR, FAOR or YSSY).	MODE shows **NAV**.
6	flyPad > Failures > ATA 34 Navigation > **GPS 1**.	GPS 1 shows FAULT, its data dashed; GPS 2 unchanged. Clear the failure: GPS 1 goes back through ACQ to NAV.

2. GPIRS position and GPS PRIMARY

> **Aircraft:** A32NX and A380X | **Where:** MCDU PROG page (A32NX); MFD POSITION / MONITOR (A380X); ND messages | **Status:** Tested in the sim by the user: OK.

What it is for

Each IR of the ADIRUs now mixes its inertial position with the position of one GPS receiver: the GPIRS position. The selection follows A320 FCOM DSC-22_20: IR 1 uses GPS 1, IR 2 uses GPS 2, IR 3 uses GPS 1 (GPS 2 with the ATT HDG selector on F/O 3). If one receiver is rejected, all the IRs use the other one. The same rule is used on the A380X (design choice: the A380 FCOM does not give it).

The FMS (the same code in the A32NX FMGC and the A380X FMC) uses the onside GPIRS, else GPIRS 3, else the offside one (A380 FCOM DSC-22-FMS-10-30-10). Its estimated position uncertainty (EPU) is the GPIRS figure of merit. Accuracy is **HIGH** when the EPU is within the required navigation performance (RNP), and **GPS PRIMARY** needs accuracy HIGH and the integrity limit within the RNP. Before, a random estimate gave GPS PRIMARY.

What you should see

Situation	What you should see
IRs aligned, receivers in NAV	GPS PRIMARY (A32NX PROG page, A380X POSITION / MONITOR), accuracy HIGH.
Receivers still in ACQ, or IRs not in NAV	No GPIRS position: no GPS PRIMARY; the FMS uses radio or IRS positions.
GPS 1 failed	IR 1 and IR 3 switch to GPS 2: GPS PRIMARY stays.
Both GPS failed	GPS PRIMARY is lost: the FMS shows GPS PRIMARY LOST.

How to use

1. Align the IRs as usual. GPS PRIMARY comes once the IRs are in NAV and the receivers are in NAV.
2. Check GPS PRIMARY on the A32NX MCDU PROG page or the A380X MFD POSITION / MONITOR page before an RNAV procedure.

How to test

#	Do this	Expected result
1	Cold and dark, power the aircraft and start the IR alignment. Open the A32NX PROG page (or the A380X POSITION / MONITOR page).	No GPS PRIMARY until the IRs are aligned and the receivers in NAV.
2	Wait for the end of the alignment.	GPS PRIMARY appears (green on the A32NX PROG page).
3	flyPad > Failures > ATA 34 Navigation > **GPS 1**.	GPS PRIMARY stays (the IRs use GPS 2).
4	Also fail **GPS 2**.	GPS PRIMARY disappears; GPS PRIMARY LOST is shown. Clear both failures: GPS PRIMARY comes back once a receiver is in NAV again.

3. GPS pages: MCDU GPS MONITOR and MFD POSITION/GPS

> **Aircraft:** A32NX (MCDU), A380X (MFD) | **Where:** A32NX MCDU DATA > GPS MONITOR; A380X MFD FMS > POSITION > GPS | **Status:** Tested in the sim by the user: OK.

What it is for

Both pages showed the sim GPS solution for both receivers with random satellite counts and figures of merit. They now show the data of each receiver.

A32NX MCDU GPS MONITOR

Per receiver (A320 FCOM DSC-22_20-50-10-28): GPS1(2) POSITION, TTRK (true track, in degrees: the old page showed the magnetic track in radians), GS, UTC, MERIT (figure of merit, FT), MODE/SAT (mode and number of satellites), GPS ALT. The data is dashed while the receiver is not in NAV (or ALTAID / AIDED), and for a receiver that no ADIRU has selected ("the data of the GPS that is not selected is dashed").

A380X MFD POSITION/GPS

Per receiver (A380 FCOM DSC-22-FMS-20-30): position, MODE (TEST, INIT, ACQ, NAV, ALTAID, AIDED, DIFF, FAULT), NBR OF SAT (while acquiring and navigating), ACCURACY (HFOM, in FT as on the FCOM figure), TRK, UTC, ALT and GND SPD. Dashes while the receiver does not compute a position.

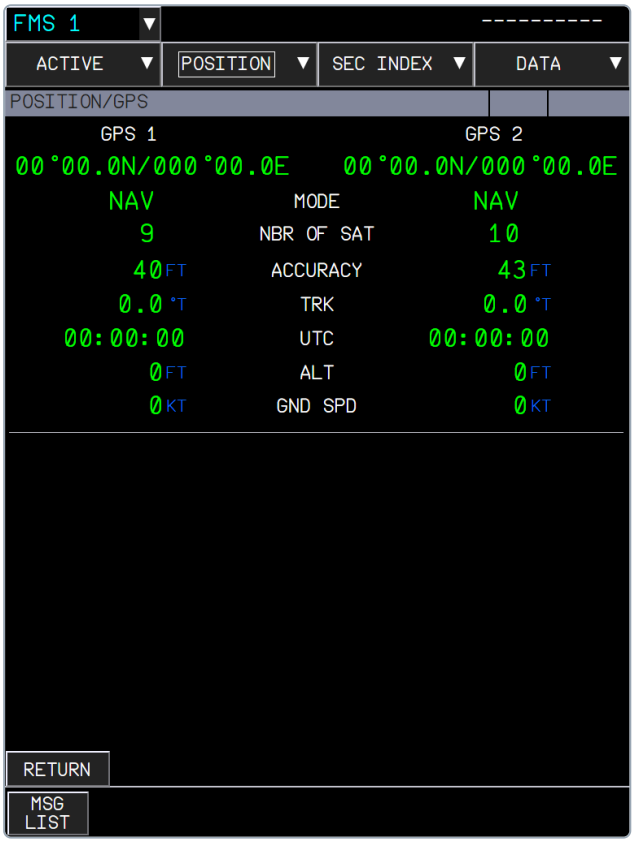


Figure 1. MFD POSITION/GPS: the page layout (picture from the browser harness, taken before the receivers existed: the values are the old stand-in values).

How to use

1. A32NX: MCDU **DATA** key > **GPS MONITOR**.
2. A380X: MFD **FMS** > **POSITION** menu > **GPS**.

How to test

#	Do this	Expected result
1	A32NX, powered, receivers in NAV: MCDU DATA > GPS MONITOR.	Two receivers with their own position, TTRK in degrees (same as the true track), GS, MERIT in FT and MODE/SAT such as NAV/9.

2	Fail GPS 1 (flyPad > Failures > ATA 34 Navigation > **GPS 1**).	GPS1 line: MODE/SAT shows FAULT, position, TTRK, GS, MERIT and GPS ALT dashed.
3	A380X, powered: MFD FMS > POSITION > GPS.	GPS 1 and GPS 2 columns with their own values; MODE NAV or DIFF.
4	Fail GPS 2.	GPS 2 column: MODE FAULT, data dashed, UTC --:--:--.

4. NAV GPS 1(2) FAULT alerts

> **Aircraft:** A32NX and A380X | **Where:** E/WD; SD STATUS page | **Status:** Tested in the sim by the user: OK.

What it is for

There was no GPS alert. A failed receiver (mode FAULT) now raises the FCOM alert:

- **A32NX: NAV GPS 1 FAULT** / **NAV GPS 2 FAULT** (FCOM PRO-ABN-NAV: "GPS 1(2) is failed"), amber caution, no procedure line. Flight phase inhibition as the radio altimeter faults (design choice). STATUS: INOP SYS GPS 1(2); with both: INOP SYS GPS 1+2 and the information FLS LIMITED TO F-APP + RAW (see "A32NX ECAM STATUS page").
- **A380X: NAV GPS 1 FAULT**, **NAV GPS 2 FAULT** or **NAV GPS 1+2 FAULT**, crew awareness (the A380 FCOM gives no procedure), inhibited as the ADR faults (design choice). STATUS (FCOM PRO-ABN-ECAM-10-34-10): INOP SYS GPS 1, GPS 2 or GPS 1+2; with both also BTV and ROW/ROP in APPR & LDG.

How to use

1. Read the alert on the E/WD and clear it with CLR.
2. Check the inoperative systems on the SD STATUS page (STS key).

How to test

#	Do this	Expected result
1	A32NX on the ground, engines off or at idle (not during the takeoff run). flyPad > Failures > ATA 34 Navigation > **GPS 1**.	E/WD: **NAV GPS 1 FAULT** in amber, MASTER CAUT.
2	Press CLR, then open the SD STATUS page.	INOP SYS: GPS 1.
3	Also fail **GPS 2**.	NAV GPS 2 FAULT; STATUS: INOP SYS GPS 1+2, and FLS LIMITED TO F-APP + RAW in green.
4	A380X on the ground: fail **GPS 1**.	E/WD: **NAV GPS 1 FAULT**. SD STATUS: INOP SYS GPS 1.
5	Also fail **GPS 2**.	The alert becomes **NAV GPS 1+2 FAULT**. STATUS: INOP SYS GPS 1+2; APPR & LDG: BTV, ROW/ROP.

5. GPS deselection and position uncertainty

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU SELECTED NAVAIDS; A380X MFD POSITION > NAVAIDS | **Status:** Tested in the sim by the user: OK on both aircraft.

What it is for

The crew can tell the FMS not to use the GPS (for example when told GPS is unreliable in an area). The A32NX DESELECT key was inoperative and the A380X had no GPS option. Now:

- **A32NX** SELECTED NAVAIDS page, key [6L] (A320 FCOM DSC-22_20-50-10-28): GPS SELECTED / DESELECTED in blue small font, the action ({DESELECT or {SELECT) in white.

- **A380X** POSITION / NAVAIDS page: GPS SELECTED / DESELECTED radio buttons (A380 FCOM DSC-22-FMS-20-30).
- With the GPS deselected the GPIRS position is no longer used: the FMS goes to IRS/DME/DME, IRS/VOR/DME or IRS only, and GPS PRIMARY is lost.
- The deselection is reset at the end of the flight (DONE phase).
- **GPS IS DESELECTED** (A32NX, white MCDU message) / **GPS DESELECTED** (A380X, white MFD message) when the GPS is deselected and the aircraft is less than 80 NM from the top of descent, or in the approach phase. The message is removed when the GPS is selected again.
- **Position uncertainty without GPS** (A320 FCOM DSC-22_20-20-20 table): IRS/DME/DME 0.27 to 0.37 NM, IRS/VOR/DME 0.30 to 0.42 NM (growing with the distance to the navaids, up to 100 NM: design choice), IRS only growing from its starting value at 6 NM/h for 40 min, 0 for 50 min, 4 NM/h for 40 min, 0 for 45 min, then 2 NM/h; 5 NM after an IRS alignment.

Limitations and why

Limitation	Why	What to do
The 0.2 NM reset of the uncertainty at takeoff is not done.	Not part of this change.	None.

How to use

1. A32NX: MCDU **DATA** > **POSITION MONITOR** > **NAVAIDS>** [6R] > press [6L] **{DESELECT**. Press [6L] **{SELECT** to use the GPS again.
2. A380X: MFD **FMS** > **POSITION** > **NAVAIDS**, GPS: select **DESELECTED** (or **SELECTED**).

How to test

#	Do this	Expected result
1	A32NX, IRs aligned, GPS PRIMARY on the PROG page. MCDU DATA > POSITION MONITOR > NAVAIDS> [6R].	Line 6 left: GPS SELECTED in blue small font, {DESELECT in white.
2	Press [6L].	GPS DESELECTED, {SELECT. The PROG page loses GPS PRIMARY; the EPU becomes a radio or IRS only value (for example 0.27 to 0.42 NM near DMEs).
3	Type something in the scratchpad, then press [6L].	NOT ALLOWED; the selection does not change.
4	In flight with a flight plan, GPS deselected, fly to less than 80 NM from the T/D.	White MCDU message **GPS IS DESELECTED**.
5	Press [6L] {SELECT.	The message is removed; GPS PRIMARY comes back.
6	A380X: MFD FMS > POSITION > NAVAIDS, GPS: DESELECTED.	POSITION / MONITOR no longer shows GPS PRIMARY. Near the T/D (80 NM) or in approach: MFD message **GPS DESELECTED**.
7	Finish the flight (DONE phase).	The GPS is selected again.

6. A32NX ECAM STATUS page and STS reminder

> **Aircraft:** A32NX | **Where:** SD STATUS page (STS key on the ECAM control panel); bottom of the E/WD | **Status:** Tested in the sim by the user: OK. On this branch the STS reminder logic was adapted to the CLR logic of FlyByWire master (the fork version sits on a fork-only commit): that adaptation is not sim-tested.

What it is for

The A32NX STATUS page was a frame that always showed NORMAL. The FWC now fills it (A320 FCOM DSC-31-20 and DSC-31-25-20):

- Left column: the information lines in green (the limitations in blue are added by `core/a32nx-air-conditioning`). Right column: **INOP SYS** (white, underlined) and the inoperative systems in amber. An empty page shows NORMAL in green.
- The page appears by itself when it is not empty and the crew clears the last alert of the E/WD with CLR, or selects CONF 1 for the approach (flaps lever 0 to 1 in flight). The next CLR (or STS) push removes it.
- **STS reminder:** STS in white, framed, at the bottom of the E/WD, while the STATUS page is not empty and not shown and no alert is shown.
- On this branch only the GPS alerts fill it (INOP SYS GPS 1, GPS 2, GPS 1+2; FLS LIMITED TO F-APP + RAW). Other branches add their alerts.

How to use

1. After clearing the alerts with CLR, read the STATUS page that comes up, then press CLR to remove it.
2. When STS shows at the bottom of the E/WD, press **STS** to review the status.

How to test

#	Do this	Expected result
1	A32NX powered, no failure. Press STS.	STATUS page: NORMAL in green.
2	Fail GPS 1 (flyPad > Failures > ATA 34 Navigation > **GPS 1**).	NAV GPS 1 FAULT on the E/WD.
3	Press CLR.	The SD shows the STATUS page by itself: INOP SYS (underlined) and GPS 1 in amber.
4	Press CLR again.	The STATUS page goes; **STS**, white and framed, shows at the bottom of the E/WD.
5	Press STS.	The STATUS page shows; the STS reminder goes off while it is shown.
6	Clear the failure.	STATUS back to NORMAL; no STS reminder.
7	In flight with GPS 1 failed and its alert cleared, select flaps 1 for the approach.	The STATUS page comes up by itself.

7. FM position: IRS position plus a GPS or radio bias, POSITION MONITOR pages

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU DATA > POSITION MONITOR, PROG, IRS MONITOR; A380X MFD FMS > POSITION > MONITOR; ND | **Status:** Not sim-tested (added; unit tests: FmPosition.spec 27, PositionUncertainty +3, PositionMonitorFields 7).

What it is for

The FMS position was the sim position. The FCOM FM position is now modelled (shared FMS code; A320 FCOM DSC-22_20-20 navigation, A380 FCOM FMS position computation):

- **FM position = MIX IRS position + a bias.** The bias follows the GPIRS position (on the ground and in flight) or the radio position (in flight only). In IRS only navigation the last bias is kept.
- Each FMS uses the GPIRS of its own side first, then ADIRU 3, then the other side.
- Without GPS PRIMARY, the position is updated at takeoff to the runway start plus the TO SHIFT.

- A manual position update (A32NX PROG **UPDATE AT**) moves the FM position; the EPU (estimated position uncertainty) follows the updates.
- The FMS writes `L:A32NX_FM_GPS_POS_DISAGREE`: the NAV FM/GPS POS DISAGREE alerts read it (their FWC and FWS parts are on `core/failures-a32nx` and `core/a380x-elec-fuel-alerts`).
- **A32NX MCDU:** POSITION MONITOR (and FROZEN) shows the real FMGC 1 and 2, GPIRS or radio, MIX IRS and IRS positions with the navigation mode and the deviations; PROG UPDATE AT works; the IRS page shows the GPIRS position and accuracy of its ADIRU.
- **A380X MFD POSITION / MONITOR:** the onside and offside FMS positions, the navigation mode, RADIO, MIXIRS, GPIRS, GPS 1 and GPS 2 from the receiver data.

Situation	What you should see
GPS PRIMARY (normal)	The FM position follows the GPIRS position; it carries the small GPS receiver error, so it differs a little from the sim position.
GPS deselected or failed, in flight, radio navaids in range	The bias follows the radio position (IRS/DME/DME or IRS/VOR/DME).
IRS only	The last bias is kept; the position drifts only as the IRS position does (see the limits).
Takeoff without GPS PRIMARY	The FM position jumps to the runway start plus the TO SHIFT at the takeoff thrust.

Limitations and why

Limitation	Why	What to do
No IRS drift and no radio position reasonableness check.	The sim ADIRS writes the sim position: there is no drift to correct.	-
No A380X POSITION UPDATE panel, no FM/IR POS DISAGREE.	Left for later.	-
No offsets, SEC / ALTN predictions, FLS beam or A32NX TRIP/EFOB table on these pages.	Left for later.	-

How to use

1. A32NX: MCDU DATA > POSITION MONITOR to compare the FMGC, GPIRS / radio, MIX IRS and IRS positions; FREEZE to hold them.
2. A32NX: for a manual update, PROG > UPDATE AT, enter the position (or a navaid ident) and confirm.
3. A380X: MFD FMS > POSITION > MONITOR.

How to test

#	Do this	Expected result
1	A32NX on the ground, IRS aligned, GPS PRIMARY. MCDU DATA > POSITION MONITOR.	FMGC1, FMGC2, GPIRS, MIX IRS and IRS 1-3 lines with real coordinates and the navigation mode; the deviations are small.
2	A380X: MFD FMS > POSITION > MONITOR.	FMS 1 and FMS 2 positions, the navigation mode, RADIO, MIXIRS, GPIRS, GPS 1 and GPS 2.
3	Both: deselect the GPS (A32NX SELECTED NAVAIDS, A380X POSITION / NAVAIDS) and take off.	At takeoff the FM position is updated to the runway start (plus the TO SHIFT); the ND symbol stays on the runway.

4	A32NX in flight: PROG > UPDATE AT a navaid ident, confirm.	The FM position moves to the update; the EPU changes.

What PR testers should check

- The FMS position now carries the GPS receiver error: watch LNAV and the ND for jumps, also when the GPS powers up.
- The takeoff update with the GPS deselected.
- The POSITION MONITOR pages of both aircraft.
- Not sim-tested yet.

8. FMS messages from the FCOM lists, and CHECK APPR SELECTION

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU scratchpad messages; A380X MFD FMS message list | **Status:** Not sim-tested (added; unit tests: FmsMessageConditions.spec and CheckApproachSelectionLogic.spec, failing first).

What it is for

Messages of the A320 MCDU message list (FCOM DSC-22_20-90) and of the A380 MFD FMS message list that the FMS never raised:

Message	Aircraft	When (FCOM)
AREA RNP IS xx.xx	both	The RNP entered is larger than the default RNP of the flight area, and the database has no RNP for the active leg.
PROCEDURE RNP IS xx.xx (A380X: PROC RNP IS)	both	The RNP entered is larger than the database RNP of the active leg.
CHECK DATA BASE CYCLE (A380X: CHECK DATABASE CYCLE)	both	Today's date is outside the effectivity of the active navigation database (an unknown effectivity raises nothing).
CHECK IRS/AIRPORT POS (A380X: CHECK IRS / ARPT POSITION)	both	An IRS alignment position is 5 NM or more from the airport reference point.
IRS ONLY NAVIGATION	A32NX	In flight, en route, in IRS only navigation for more than 10 min, or at the start of the descent.
CHECK DEST DATA	A32NX	The destination data were received by uplink (wind not entered by the crew).
CRZ FL ABOVE MAX FL	A32NX	The crew enters a cruise level above the computed maximum level.
TIME TO EXIT	both	In the holding, the extra fuel is zero or less.
EXPECT TURN AREA EXCEEDANCE	A380X	The active or next leg is a procedure turn that exceeds the excursion distance of the database.
SPECIF NDB NOT AVAIL	A380X	The NDB of the NDB approach is deselected.
T.O TIME REACHED	A380X	The clock reaches the ETT; shown until the takeoff or a new ETT.
ADJUST DESIRED TRK OR HDG	A380X	A DIR TO with CRS IN or CRS OUT is inserted.
CHECK APPR SELECTION	A32NX	A non ILS approach in the F-PLN and an ILS tuned manually on the RAD NAV page, in cruise less than 100 NM from the T/D, in descent or in approach. Amber; it also sets the variable the PFD FMA line 3 shows.

The messages are type II: sent once when the condition starts and removed when it ends. A message with a value (AREA RNP IS 2.00) is replaced when the value changes.

Limitations and why

Limitation	Why	What to do
Other messages of the FCOM lists are still missing.	They need data or functions the FMS does not have yet.	-
CHECK APPR SELECTION on the PFD FMA needs `core/failures-a32nx`.	The PFD FMA part of the change is on that branch; here only the MCDU message and the variable.	Merge both.
The A380X MFD System Message Area (CHECK FMS MESSAGE) is on `core/a380x-atccom-datalink`.	It also shows the ATIS of the ATC COM datalink of that branch.	-

How to use

1. Read the message in the MCDU scratchpad (A32NX; CLR removes it) or in the MFD FMS message list (A380X).
2. Act as the FCOM says for each message (check the RNP, the database cycle, the IRS position, the approach selection...).

How to test

#	Do this	Expected result
1	A32NX with a current navigation database, cold and dark to cruise.	No CHECK DATA BASE CYCLE at any time.
2	A32NX in cruise less than 100 NM from the T/D, F-PLN with an RNAV (non ILS) approach. RAD NAV: type an ILS frequency in LS.	MCDU: **CHECK APPR SELECTION** in amber.
3	A32NX in flight, en route: PROG page, type 2.0 in the REQUIRED (RNP) field.	**AREA RNP IS 2.00** (or PROCEDURE RNP IS on a procedure leg with its own RNP).
4	A32NX: enter a CRZ FL above the MAX REC level on PROG.	**CRZ FL ABOVE MAX FL**.
5	A380X: F-PLN > DIR TO with CRS IN, INSERT.	FMS message **ADJUST DESIRED TRK OR HDG**.
6	A380X on the ground: F-PLN, vertical revision of the origin, enter an ETT a minute ahead and wait.	**T.O TIME REACHED** at that time, gone at the takeoff.

What PR testers should check

- CHECK DATA BASE CYCLE stays silent with a current database (the main risk of a wrong date check).
- Any of the new messages showing when the FCOM condition is not met.
- Not sim-tested yet.